

PUBLIC COMMENT & EVIDENCE DOSSIER

Targeted Oral Naltrexone for Alcohol Use Disorder

A high-priority Scenario 1 nomination for systematic FDA review

Proposed regimen

**Naltrexone 50 mg orally approximately 60 minutes before anticipated drinking;
maximum 50 mg in 24 hours, with comprehensive AUD management**

Docket	FDA-2026-N-4492
RFI	Drug Repurposing for Unmet Medical Needs
Submitted by	Curada Public Benefit Corporation
Point of contact	Mariia Markitanova, Founder & Director
Submission date	July 13, 2026
Scope	Same 50-mg oral dosage form and route; new event-driven regimen within the approved alcohol-dependence disease area

Requested FDA action

Conduct a systematic review of the public evidence; determine whether it satisfies the substantial-evidence standard for the proposed condition of use; initiate the appropriate MODERN or literature-led labeling process if it does; and, if a focused gap remains, specify a confirmatory or real-world evidence pathway rather than allowing this generic-drug opportunity to remain stranded.

This submission distinguishes established facts, supportive precedent, biologic hypotheses, provider-reported reach, and unresolved evidence gaps.

Contents

Executive summary	2
1. Nomination and fit with FDA’s RFI	3
2. Regulatory pathway: ReVia, MODERN, and literature-led precedent	4
3. Mechanism and dosing rationale	5
4. Research history and the expired method patent	6
5. Clinical evidence	7
6. Safety, patient selection, and risk controls	8
7. United Kingdom and European precedent	10
8. U.S. practice, patient demand, and the labeling gap	11
9. Prospective real-world evidence package	12
10. Proposed labeling concept for FDA evaluation	13
11. Other reward-related disorders: hypothesis, not this nomination	14
12. Limitations and evidence-integrity commitments	14
13. Specific requests to FDA	15
References and source links	16
Appendix A. Web comment (under 500 words)	18

Executive summary

Bottom line

Targeted oral naltrexone is a credible, high-public-health-value Scenario 1 candidate, but the record should be reviewed—not overstated. The same 50-mg oral product is already approved for alcohol dependence; the proposed change is timing and frequency. Several randomized studies directly test event-driven use, while the broader daily-dose literature anchors the known AUD benefit–risk profile. Trial heterogeneity means FDA should make the substantial-evidence determination after a transparent systematic review and identify any narrowly tailored residual evidence need.

The case in six points

- Fit with the RFI. The nomination concerns an approved generic drug, the same dosage form and route, a major unmet need, a mature safety database, and little apparent commercial incentive for a sponsor-led supplement. [1–5]
- Regulatory opening. FDA determined in April 2026 that ReVia (NDA 018932) was not withdrawn for safety or effectiveness. That finding preserves generic pathways and appears to remove one key barrier to MODERN eligibility; FDA must confirm all statutory predicates. [3–5]
- Direct evidence. Event-driven studies span an early feasibility study, randomized trials in problem drinkers and alcohol dependence, and a 2022 placebo-controlled trial with positive prespecified drinking outcomes in a selected population. Results are promising but not uniform. [11–17]
- Mechanistic plausibility. Pre-exposure opioid antagonism can reduce alcohol-related endogenous-opioid reward and downstream mesolimbic signaling. Repeated reduced reinforcement is often called pharmacological extinction. The exact mechanism in alcoholism remains unresolved, and the phrase should not substitute for clinical evidence. [6–10, 52]
- External precedent. EMA-authorized, NICE-recommended as-needed nalmefene validates the clinical and regulatory plausibility of pre-drinking opioid-system modulation, but it is not direct proof for naltrexone and NICE does not endorse targeted naltrexone. [20–23]
- Actionable evidence pathway. Existing digital care infrastructure makes a prospective, auditable registry or pragmatic study feasible now. Such evidence should complement—not replace—the statutory effectiveness standard. [26–32]

Requested determination

1. Place targeted oral naltrexone for AUD on FDA’s high-priority Scenario 1 review list.
2. Commission or coordinate a protocol-driven systematic review and, where feasible, individual-participant or harmonized reanalysis of the targeted trials.
3. If the record supports approval, initiate the appropriate §503D MODERN process and/or publish a Federal Register notice and scientific assessment that can support labeling submissions.
4. If the record falls narrowly short, state the specific gap and offer an efficient Scenario 2 pathway using a pragmatic randomized trial and prospective real-world evidence.
5. Convene FDA, generic manufacturers, researchers, clinicians, patients, and Curada or other capable nontraditional developers to operationalize the pathway.

Important boundary

This dossier does not claim that: a patent proves efficacy; the April 2026 ReVia notice validates the targeted regimen; NICE recommends targeted naltrexone; nalmefene evidence is interchangeable with naltrexone evidence; targeted dosing is approved in Australia; or current provider/community use establishes effectiveness. Those distinctions protect the credibility of the nomination.

1. Nomination and fit with FDA’s RFI

Nomination. Curada nominates naltrexone hydrochloride 50-mg oral tablets for adults with AUD whose treatment goal includes reducing heavy drinking, used approximately 60 minutes before anticipated alcohol consumption and no more than once in 24 hours, as part of comprehensive management. This is a proposed labeling concept for FDA review—not a prescribing recommendation or assertion of current approval.

RFI criteria crosswalk

RFI consideration	Record for this nomination	Assessment
Approved drug / new use	Oral naltrexone is approved for alcohol dependence. The proposed change is an event-driven regimen within that disease area.	Strong fit
Compelling effectiveness evidence	Multiple direct targeted studies plus a large daily-dose literature; direct findings are promising but heterogeneous.	Plausible Scenario 1; FDA review needed
Same dosage form and route	Same 50-mg tablet and oral route; no new formulation or device is required.	Meets
Comparable safety profile	Exposure is intermittent rather than greater than approved daily exposure, but timing around drinking and selected populations require explicit safeguards.	Likely manageable; confirm from full record
No apparent commercial interest	Generic competition, expired method patent, withdrawn RLD approval, and no remaining listed patents/exclusivity create weak private incentives.	Strong fit
Public-health value	AUD is common, medication uptake is exceptionally low, and alcohol-related mortality is substantial.	Strong fit

Why the unmet need is material

In 2024, an estimated 28.0 million people aged 12 or older in the United States had past-year AUD; only 7.6% received any alcohol treatment and 2.5% received medication-assisted treatment. [24] CDC estimated roughly 178,000 annual deaths from excessive alcohol use during 2020–2021. [25] These figures do not prove that one regimen will close the treatment gap, but they establish the scale of the public-health opportunity and the value of an additional labeled option for patients pursuing drinking reduction rather than immediate abstinence.

Scenario 1—strong candidate, not a foregone conclusion

The record supports high-priority Scenario 1 review. The RFI contemplates candidates for which sufficient evidence may already exist and notes that one adequate and well-controlled investigation plus confirmatory evidence can sometimes constitute substantial evidence. [1] The targeted-naltrexone record contains more than one randomized investigation, a convergent daily-dose evidence base, a mature safety record, and a class/regimen precedent. Yet different populations, targeting rules, timing instructions, counseling cointerventions, and outcome patterns prevent a responsible commenter from declaring the legal standard satisfied without FDA’s full review. If FDA finds a narrow residual gap, the candidate should transition to an efficient evidence-generation track—not be excluded from the initiative.

2. Regulatory pathway: ReVia, MODERN, and literature-led precedent

April 2026 ReVia determination

FDA's April 16, 2026 notice states that ReVia 50-mg oral tablets (NDA 018932), approved in 1984, were withdrawn at the holder's request effective May 1, 2019, and were not withdrawn for reasons of safety or effectiveness after FDA reviewed agency records, published literature, and postmarketing adverse-event data. [3] The determination allows FDA to continue approving abbreviated applications that reference ReVia, subject to applicable requirements. It does not determine that targeted dosing is safe or effective; it is relevant because it confirms the regulatory status of the RLD and the enduring acceptability of generic reliance on it.

Correct MODERN analysis

The relevant provision is FD&C Act §503D, 21 U.S.C. §353d—not §505-1 or §505(o)(4). [4] For a “covered drug,” FDA may initiate a labeling-update process when the statutory predicates are met and FDA determines that an update would benefit public health. The condition-of-use clauses are alternatives: new scientific evidence concerning a new or existing condition of use; labeling that does not reflect current legal/content/format requirements; or a relevant accepted use not reflected in labeling. A “relevant accepted use” must be supported by evidence that appears to FDA to meet §505 approval standards; popularity alone is insufficient.

§503D element	Application to ReVia / targeted naltrexone	Caveat
NDA approved under §505(c)	ReVia NDA 018932 was an approved NDA.	FDA to confirm statutory record.
No unexpired listed patent/exclusivity	The historical method patent expired; current Orange Book data appear to show no remaining listed protection.	FDA should confirm as of initiation date.
Approval withdrawn for reasons other than safety/effectiveness	FDA made this determination in April 2026.	This finding concerns ReVia, not the proposed regimen.
Public-health benefit	Potential to broaden evidence-based treatment choice in a large, undertreated population.	Benefit depends on evidence and implementable safeguards.
New scientific evidence—prong I	Trials and later evidence are not reflected in the historic ReVia label.	Evidence must meet the applicable approval standard.
Relevant accepted use—prong III	Event-driven use is practiced and discussed in clinical education.	FDA—not utilization volume—decides whether evidence appears to meet §505.

A complementary literature-led pathway

FDA has previously performed systematic literature analyses, published findings, and encouraged sponsor submissions. For inhalational anthrax, FDA reviewed literature and adverse-event data and explained how applicants could rely on its scientific publication set. [40–41] For radionuclide contamination, FDA reviewed literature and observational evidence and invited 505(b)(2) applications for pentetate calcium trisodium and pentetate zinc trisodium. [42–44] Most recently, FDA approved a Wellcovorin indication after systematically reviewing case-level and mechanistic literature in collaboration with the NDA holder. [56] These are not automatic label changes and differ from AUD, but they demonstrate a current agency-led evidence-curation model that can coexist with MODERN while preserving approval standards.

The incentive failure is structural

Duke-Margolis identifies a generic-repurposing market failure: low prices and competition erode returns on sponsor-funded development, leaving off-label uses with uneven uptake, reimbursement, awareness, supply, and liability barriers. [26] Because an updated use would benefit every generic competitor, the funder may be unable to capture the return—a classic free-rider problem. Patent expiry is not efficacy evidence; it explains why a valuable use can remain stranded despite a public record.

3. Mechanism and dosing rationale

Concise, defensible mechanism statement

Naltrexone is a competitive, nonselective opioid-receptor antagonist. Administered before alcohol exposure, it can attenuate endogenous-opioid reward signaling—especially μ -opioid-receptor effects—and downstream mesolimbic dopamine responses. Repeated pairing of alcohol-related cues and consumption with reduced reinforcement may weaken learned reward value. This learning model is commonly called pharmacological extinction; FDA-approved labeling states that naltrexone's mechanism in alcoholism is not fully understood.

Stepwise model

1. Alcohol exposure can activate endogenous-opioid pathways, including β -endorphin signaling, that contribute to reward-related mesolimbic dopamine activity. [7–10]
2. A pre-exposure opioid antagonist reduces part of that reinforcing signal; it does not prevent intoxication, eliminate impairment, or make drinking safe.
3. Across repeated cue-behavior episodes, reduced reinforcement may weaken learned alcohol reward and cue-driven motivation. “Extinction” is a model of this gradual learning process, not a guaranteed response to each dose.
4. Because reward learning is behavioral and contextual, adherence to pre-drinking timing and accompanying clinical support matter. The intervention is not simply “take a pill whenever convenient.”

Why approximately 60 minutes is plausible—but not proven optimal

Current oral labeling reports peak plasma concentrations of naltrexone and 6β -naltrexol within about one hour, with approximate half-lives of 4 and 13 hours, respectively; 50 mg can block opioid effects for at least 24 hours. [6] These pharmacokinetics support pre-exposure administration, and direct trials have commonly instructed use about 1–2 hours before a high-risk situation. They do not establish that exactly 60 minutes is superior to 45, 90, or 120 minutes. FDA should review timing as an estimand and consider “approximately 60 minutes (preferably 1–2 hours)” if supported by the pooled evidence.

Human proof of pharmacology

In an 18-participant randomized human-laboratory study, six days of 50-mg oral naltrexone reduced craving and subsequent alcohol self-administration after a priming drink; participants drank fewer drinks and drank more slowly than with placebo. [52] Because treatment was scheduled daily rather than event-driven, this study supports attenuation of alcohol reinforcement but does not establish the proposed timing instruction or the effectiveness of a targeted regimen.

What should not be claimed

- The exact molecular mechanism of naltrexone in AUD is settled. It is not; the approved label says it is not understood.
- Targeted dosing has proven a tolerability or dropout advantage over daily dosing. Lower pill burden is plausible, but comparative evidence is insufficient.
- Off-days fully restore opioid-mediated reward. A 50-mg dose can block opioid effects beyond the immediate drinking episode.
- Daily naltrexone is obsolete or biologically inferior. Daily oral naltrexone remains evidence-based and approved; targeted use would be an additional option.
- The regimen prevents withdrawal. Patients at risk of severe alcohol withdrawal require medical assessment and, when indicated, supervised withdrawal management.

4. Research history and the expired method patent

A research program, not a single-origin claim

The evidence should not be framed as resting on one researcher. John D. Sinclair’s Finnish program connected animal learning studies from the 1970s with later human investigations and synthesis, but the modern clinical record includes independent teams, multiple countries, randomized designs, and contemporary investigators. [10–17] A careful formulation is: “a Finnish research program spanning animal work from the 1970s through human trials around 2000, followed by a broader international clinical literature.” It would be inaccurate to say the human regimen has been clinically proven since 1972.

U.S. Patent 4,882,335

The patent “Method for treating alcohol-drinking response,” filed June 13, 1988 and issued November 21, 1989, claims repeated alcohol consumption while an opiate antagonist—including naltrexone—blocks reinforcement, with extinction as the intended result. [19] It describes dosing “shortly before” drinking and a preferred naltrexone range of 50–200 mg; it does not itself establish an exact 60-minute interval. The patent is expired. Its evidentiary relevance is historical: the concept was sufficiently defined to be claimed decades ago. Its economic relevance is the absence of continuing method-patent exclusivity. A patent is not peer review, regulatory endorsement, or proof of clinical benefit.

Milestone	What it supports	What it does not establish
1970s Finnish animal research	Longstanding learning/reinforcement hypothesis.	Human effectiveness or a label-ready regimen.
1988 filing / 1989 issue of U.S. 4,882,335	Early, specific articulation of drinking-under-blockade method.	FDA validation or a 60-minute optimum.
1997–2009 targeted clinical studies	Feasibility and randomized human evidence across several designs.	Uniform benefit in all populations.
2022 event-driven RCT	Contemporary placebo-controlled efficacy signal with high completion.	Broad generalizability beyond the selected study population.
2026 ReVia finding	RLD was not withdrawn for safety/effectiveness.	Approval of event-driven dosing.

5. Clinical evidence

Evidence framework. The most probative evidence directly randomizes an event-driven naltrexone schedule against placebo and prespecifies drinking outcomes. Daily-dose trials support the drug's general AUD benefit-risk profile but cannot, by themselves, establish the new regimen. Mechanistic studies and European nalmefene experience are supportive, not substitutive.

Direct targeted or event-driven studies

Study	Design / regimen	Principal finding	Interpretive limit
Kranzler 1997 [11]	n=21 uncontrolled feasibility; 50 mg before high-risk situations, 2–5 times/week.	Drinking outcomes improved; feasibility supported.	No placebo control; authors called for a trial.
Heinälä 2001 [12]	n=121, 2×2 drug × psychotherapy; 50 mg daily for 12 weeks, then targeted for 20 weeks.	Coping-skills + naltrexone group showed lower heavy-drinking relapse; no corresponding supportive-therapy benefit.	Targeted phase followed daily lead-in; cannot isolate de novo targeting.
Kranzler 2003 [13] / reanalysis [14]	n=153, 8-week factorial daily vs targeted and naltrexone vs placebo; 50 mg.	Targeting reduced some drinking outcomes; active-drug pattern was mixed and daily naltrexone modestly favored in primary report.	Tablet-allotment and adherence effects; no uniform targeted superiority.
Kranzler 2009 [15]	n=163, 12-week 2×2 RCT; targeted 50 mg 1–2 h before high-risk situations, 3–5 tablets/week; brief counseling.	Overall targeted contrast on primary drinks/day did not reach significance; male subgroup and exploratory drinks/drinking-day favored targeted naltrexone; 85% completion.	Sex moderation and exploratory outcome; not a broad positive primary result.
O'Malley 2015 [16]	n=128 young adults; 25 mg daily + 25 mg targeted ≥2 h before drinking.	Primary heavy-drinking and abstinence endpoints not significant; secondary intensity/BAC outcomes improved.	Not a pure 50-mg event-driven regimen.
Santos 2022 [17]	n=120 sexual- and gender-minority men with binge drinking and mild-moderate AUD; 50 mg event-driven vs placebo + weekly counseling for 12 weeks.	Binge-drinking days IRR 0.74; weeks with any binge IRR 0.83; drinks/month IRR 0.69; 93% completion; some persistence at 6 months.	Selected male population; counseling cointervention; broader replication needed.

Supportive daily-dose evidence

COMBINE randomized 1,383 participants and tested naltrexone 100 mg daily for 16 weeks, not event-driven treatment. [18] Naltrexone improved several drinking outcomes, particularly with medical management, and remains important confirmatory evidence that opioid antagonism can benefit AUD. A 2023 systematic review of 118 trials involving 20,976 participants found moderate-strength evidence for 50-mg/day oral naltrexone on several alcohol outcomes; nausea and vomiting were increased. [45] These data anchor the general therapeutic concept and known risks but cannot establish the proposed timing instruction.

A Cochrane review of 50 randomized trials (7,793 participants) found that naltrexone reduced the risk of heavy drinking versus placebo (RR 0.83, 95% CI 0.76–0.90). [53] That quantitative result is strong molecule-level support, not proof of the exact event-driven schedule. Conversely, Palpacuer's controlled-drinking network meta-analysis found no high-grade evidence and did not demonstrate a clear naltrexone effect on total alcohol consumption. [54] Candidly including that counterevidence is essential: the total record justifies a disciplined FDA review, not a declaration that every regimen-specific question is resolved.

Evidence hierarchy for FDA's decision

Evidence layer	Contribution	Cannot establish alone
Direct event-driven RCTs [12–17]	Regimen-specific efficacy, adherence, and safety under tested conditions.	A uniform 60-minute instruction or broad generalizability.

Evidence layer	Contribution	Cannot establish alone
Daily-dose trials and reviews [18, 45, 53–54]	Known AUD activity, benefit–risk context, and the full uncertainty range.	Effectiveness of intermittent pre-exposure dosing.
Human laboratory / mechanistic evidence [6–10, 52]	Biologic plausibility and attenuation of alcohol reinforcement.	Clinical durability, patient selection, or optimal timing.
EMA/NICE nalmefene precedent [20–23]	Regulatory feasibility of as-needed opioid modulation for reduction goals.	Naltrexone efficacy or pharmacologic interchangeability.
Practice and prospective RWE [26–39]	Feasibility, demand, implementation, and confirmatory evidence generation.	Substantial evidence without a fit-for-purpose design.

Integrated assessment of effectiveness

Why Scenario 1 review is warranted

The record contains randomized, placebo-controlled, regimen-specific evidence; a strong recent direct trial; longitudinal and mechanistic support; extensive daily-dose confirmation; a same-route/same-strength product; and an external as-needed opioid-modulator precedent. The question is no longer whether evidence exists, but whether the totality—after protocol-level extraction and bias assessment—meets §505.

Why FDA should resist overstatement

The studies use different targeting instructions, include counseling or daily lead-ins, enroll different severity groups, and do not uniformly meet primary endpoints. Evidence is sparse in women, severe AUD, medically complex patients, and people at withdrawal risk. “Pharmacological extinction” is rarely measured as a durable mechanistic endpoint. These are reasons for disciplined review and possibly focused confirmation—not reasons to ignore the candidate.

6. Safety, patient selection, and risk controls

Known product, new use conditions. The proposed maximum—50 mg in 24 hours—does not exceed the approved daily tablet exposure, but an event-driven regimen changes who may start treatment, how timing is communicated, and the circumstances of use. FDA should retain all product-class safeguards and add regimen-specific counseling.

Risk / issue	Proposed safeguard
Current opioid use or physiologic opioid dependence	Contraindicate consistent with labeling; verify opioid-free interval and use clinical assessment or testing when indicated. Warn of precipitated withdrawal.
Need for opioid analgesia	Provide a medication card/alert and emergency pain-management instructions; counsel that attempts to override blockade can be fatal.
Hepatic injury	Use current warnings; assess history and baseline liver tests when clinically indicated; stop and evaluate symptoms of acute hepatitis.
Alcohol withdrawal	Screen before reduction. Patients at risk of severe or complicated withdrawal need medically supervised management; naltrexone is not a withdrawal treatment.
Pregnancy / lactation and comorbidity	Retain current label precautions and individualized benefit–risk assessment.
Intoxication and impairment	State explicitly that naltrexone does not prevent intoxication, impaired driving, poisoning, injury, or other acute alcohol harms.

Risk / issue	Proposed safeguard
Adverse effects / adherence	Monitor nausea, vomiting, headache, dizziness, fatigue, mood and suicidality as clinically appropriate; capture dose timing and missed pre-dosing.
Dosing after drinking has begun	Do not imply retroactive protection. FDA should determine whether “take as soon as possible” is supported; until then, the nomination is pre-exposure use.

Safety interpretation

FDA’s 2026 ReVia determination reinforces that the original product was not withdrawn because of safety or effectiveness concerns. [3] It should be cited together with, not in place of, the current labeling and the regimen-specific trial safety data. Targeted exposure could reduce pill burden, but available studies do not establish a clinically important tolerability or discontinuation advantage versus daily dosing. A truthful label should offer choice without disparaging daily naltrexone.

7. United Kingdom and European precedent—accurately stated

Nalmefene: a class-and-regimen precedent

EMA authorized Selincro (nalmefene) across the European Union in 2013 for reducing alcohol consumption in adults with alcohol dependence who have a high drinking-risk level, without physical withdrawal symptoms and who do not require immediate detoxification, together with continuous psychosocial support. [20–21] The product is taken as needed, preferably 1–2 hours before anticipated drinking and at most once daily. Nalmefene is an opioid-system modulator (μ/δ antagonist and κ partial agonist), not an opioid agonist simpliciter.

NICE Technology Appraisal 325 recommends nalmefene within its marketing authorization for eligible adults, with psychosocial support. [22] This acknowledges that a pre-drinking, as-needed opioid-modulation strategy can be regulated, implemented, and health-technology assessed. It is persuasive precedent for modality and patient selection, not bridging evidence sufficient to approve naltrexone.

What NICE says about naltrexone

NICE Clinical Guideline 115 recommends oral naltrexone after assisted withdrawal, starting at 25 mg and increasing to 50 mg daily, normally for up to six months. [23] It does not endorse the Sinclair Method or targeted pre-drinking naltrexone. The dossier therefore cites TA325 for the nalmefene precedent and CG115 to make the distinction explicit.

Australia

Australia has registered 50-mg oral naltrexone for alcohol dependence, while national clinical guidance describes daily dosing after withdrawal resolution. [46–47] No official Australian authorization of targeted pre-drinking naltrexone was identified for this submission. Australian clinical education discussing targeted use may demonstrate awareness, but should not be described as regulatory approval.

Credibility rule

Use the European record to show feasibility of an as-needed opioid-modulator label. Do not collapse nalmefene into naltrexone, call nalmefene an opioid agonist, or represent NICE TA325 as a targeted-naltrexone recommendation.

8. U.S. practice, patient demand, and the labeling gap

A growing ecosystem supports event-driven naltrexone, but its evidence value must be classified correctly. Practice patterns and patient communities demonstrate feasibility, demand, and the possibility of prospective data capture. They do not substitute for randomized evidence or establish an accepted use under §503D unless FDA concludes the supporting evidence appears to meet §505 standards.

Organization / channel	Verified role or public statement	How it may support an FDA pathway
Options Save Lives	501(c)(3), formerly C Three Foundation; education and provider resources supporting the Sinclair Method.	Stakeholder engagement, patient education, provider-network outreach. [27]
Thrive Alcohol Recovery	Coaching, community, and referrals explicitly oriented to the Sinclair Method; not represented here as the prescriber.	Behavioral-support protocol and prospective engagement. [28]
Oar Health	Public page describes taking naltrexone before drinking and reports 35,000+ members; figure is company-reported and not a targeted-only, audited count.	Large-scale feasibility signal; potential structured data partner if data provenance is verified. [29]
Ria Health	Offers medication, coaching, app-based support and breath-alcohol tracking; says naltrexone may be taken daily or about one hour before drinking.	Existing infrastructure for dose, drinking, outcome and safety capture. [30–31]
Choose Your Horizon	Telehealth AUD treatment and coaching; public reach claims are commercial and do not clearly isolate targeted dosing.	Potential recruitment and implementation stakeholder, subject to protocol verification. [32]
Sunnyside	Educates about the Sinclair Method but states its own protocol emphasizes daily adherence.	Shows public awareness; should not be described as a targeted prescriber. [33]
Peer communities	Active Reddit/Facebook groups exchange experience and implementation advice; exact current membership was not independently verified.	Patient-priority research, recruitment and qualitative implementation work—not efficacy proof.

Provider-reported scale: useful, but not a patient count

Commercial and nonprofit sources report reach in the thousands or tens of thousands. These figures may overlap, may include daily regimens, and are not independently audited. They should be used only to show that a multi-site prospective evidence package is operationally feasible. This submission does not claim that Curada currently treats those patients or possesses an existing registry.

Media and clinical education

Targeted naltrexone has reached the public through Gabrielle Glaser’s 2015 Atlantic feature and later educational coverage, including a Harvard Health explainer. [34–36] Partnership to End Addiction also publishes a patient-oriented explanation. [37] Media attention supports awareness and the need for accurate labeling; it is not clinical evidence.

9. Prospective real-world evidence package

Curada's offer

Curada is eager to assist FDA with a transparent literature analysis and, in collaboration with clinicians, patients, researchers, generic manufacturers, and care platforms, a prospective evidence protocol. Curada does not ask FDA to treat an uncontrolled user count as proof; it offers to help convert real-world feasibility into prespecified, auditable evidence.

A fit-for-purpose protocol

Domain	Minimum specification
Population	Adults meeting validated AUD criteria, stratified by sex, severity, treatment goal, baseline heavy-drinking frequency, liver status, opioid exposure, and withdrawal risk. Exclude or separately manage those needing acute detoxification.
Exposure	50-mg oral naltrexone event-driven. Capture prescribed instruction, dose timestamp, first-drink timestamp, latency, drinking that began without pre-dose, and total tablets. Verify dispensing and product identity.
Comparator	Pragmatic randomization where feasible: targeted vs daily naltrexone, or targeted naltrexone vs matched usual care. If observational, emulate a target trial and prespecify confounding control.
Primary outcome	For adults with moderate-to-severe AUD within the qualified context of use, a two-level reduction from baseline in WHO alcohol risk-drinking level—FDA-qualified in 2025 as an acceptable primary endpoint. Otherwise, prespecify an FDA-agreed endpoint such as percent heavy-drinking days or no heavy-drinking days. [55]
Secondary outcomes	One-level WHO risk shift, drinks per drinking day, any binge week, abstinent days, AUDIT-C, craving, functioning/quality of life, treatment retention, emergency care, and patient-defined goal attainment.
Objective measures	Breath-alcohol data where acceptable; PEth and/or EtG/EtS in a subset; medication-event monitoring where feasible. State what each measure can and cannot verify.
Safety	Adverse events, serious events, opioid-related events, precipitated withdrawal, liver-related events/labs, alcohol withdrawal, overdose, injury, pregnancy outcomes where relevant, and discontinuations.
Analysis	Public protocol and statistical analysis plan; intention-to-treat estimand; per-protocol timing analysis; treatment-effect heterogeneity; sensitivity analyses for missingness, self-report error, adherence and unmeasured confounding.
Governance	IRB review as applicable, informed consent/authorization, HIPAA-compliant data handling, common data model, independent safety oversight, audit trail, conflict-of-interest disclosure, and prospective registration.
FDA interaction	Early Type C or other appropriate meeting to agree context of use, endpoint hierarchy, data standards, and how RWE will complement existing trials.

Highest-value analyses of existing evidence

- A PRISMA-conformant systematic review with published protocol, complete search strings, duplicate screening, risk-of-bias assessment, and GRADE-style certainty evaluation.
- Harmonization of “targeted” exposure: dose strength, allowed weekly frequency, anticipated high-risk situation versus anticipated drinking, 60-minute versus 1–2-hour instruction, daily lead-in, and rescue dosing.
- Outcome harmonization using FDA-relevant endpoints: qualified two-level WHO risk-drinking reduction where applicable, no heavy-drinking days, percent heavy-drinking days, drinks per drinking day, return to heavy drinking, and patient functioning. [55]

- Individual-participant data meta-analysis if investigators can share data, with prespecified sex, severity, goal, counseling and adherence interactions.
- Joint safety analysis using event-driven trials, current oral-naltrexone labeling, the daily-dose AUD literature, and postmarketing data, with special attention to opioid exposure and withdrawal screening.

10. Proposed labeling concept for FDA evaluation

The following is decision-focused draft language, not an assertion that current evidence already supports each clause. FDA should refine the population, endpoint, timing window, counseling language, and safety instructions based on its review.

Section 2 — Dosage and Administration (concept draft)

For selected adults with alcohol use disorder whose treatment goal includes reduction of heavy drinking and who do not require immediate medically supervised alcohol withdrawal, the recommended event-driven dose is 50 mg orally approximately 60 minutes before anticipated alcohol consumption. Do not take more than 50 mg in any 24-hour period. Treatment should be used as part of a comprehensive management program that includes clinical follow-up and psychosocial support. Screen for current opioid use, opioid dependence, need for opioid analgesia, hepatic risk, pregnancy as applicable, and risk of severe alcohol withdrawal before initiation. Naltrexone does not prevent intoxication or impairment and does not make alcohol consumption safe.

Decision points FDA should resolve

Label element	Evidence question
Population	Mild-moderate AUD only, or broader AUD excluding immediate detoxification? Is evidence sufficient across sex and severity?
Treatment goal	Reduction of heavy drinking, prevention of return to heavy drinking, or both? Which endpoint best maps to the evidence?
Timing	Exactly 60 minutes, approximately 60 minutes, or preferably 1-2 hours before drinking?
Frequency	Maximum once daily; should a weekly tablet range be stated or avoided?
Already drinking	Is there evidence to support administration as soon as possible after drinking begins, as with nalmefene, or only before exposure?
Counseling	What minimum psychosocial or medical-management component is supported and implementable?
Duration	Initial evaluation point, continuation criteria, and maximum studied duration.
Safety monitoring	Baseline and follow-up liver tests by risk; withdrawal screening tool; opioid-free interval and emergency analgesia language.

Position relative to approved daily dosing

Any label expansion should preserve daily oral naltrexone as a valid evidence-based option. Event-driven treatment may appeal to some reduction-goal patients and align medication with anticipated high-risk occasions. It should be offered through shared decision-making, not framed as universally superior or as a reason to stop an effective daily regimen.

11. Other reward-related disorders: hypothesis, not this nomination

Opioid signaling contributes to several reward and compulsivity phenotypes, but “endorphin-mediated disorders” is not a validated regulatory class. Evidence must be evaluated separately by indication, formulation, schedule, and endpoint. Extending the AUD learning model wholesale would weaken this submission.

Area	Current signal	Regulatory conclusion for this dossier
Gambling disorder	A randomized study found benefit with daily naltrexone across 50–150 mg doses. [48]	Supports opioid-system relevance; not evidence for pre-event dosing or AUD labeling.
Compulsive sexual behavior	Small open-label evidence and a later randomized study with mixed measures/high placebo response. [49–50]	Preliminary, daily dosing, indication-specific work required.
Binge eating / high-sugar or high-carbohydrate intake	Early naltrexone studies were small and weak; modern studies often use naltrexone/bupropion, which cannot isolate naltrexone.	No claim of established targeted benefit; separate research question.
Cocaine use disorder	Current labeling states naltrexone has not been shown to affect cocaine or other non-opioid drug use; a 50-mg/day trial did not show key benefits. [6, 51]	Do not cite as a positive extension.

Recommended scope statement

FDA may consider opioid modulation in other compulsive/reward disorders through separate RFI nominations, but those hypotheses neither add to nor subtract from the evidence for targeted naltrexone in AUD. Curada’s present request is confined to AUD.

12. Limitations and evidence-integrity commitments

- No exhaustive database search or formal risk-of-bias review was completed for this public comment; FDA’s systematic review should be protocolized and reproducible.
- Several direct studies are small, use different instructions, or report subgroup/exploratory findings; conclusions should privilege prespecified endpoints and multiplicity controls.
- The 2022 trial is strong direct evidence but enrolled a selected population of sexual- and gender-minority men; confirmation in women and broader clinical populations is important.
- Provider-reported membership and treatment counts may overlap and are not independently audited. They support feasibility, not effectiveness or prevalence.
- The exact one-hour timing instruction is pharmacokinetically plausible and commonly used, but an optimal timing window has not been established head-to-head.
- Nalmefene differs pharmacologically from naltrexone. Its EU authorization is a modality precedent, not a scientific bridge sufficient for approval.
- Current FDA labeling, not this dossier, controls clinical use. Any new use should proceed through the appropriate regulatory process.

13. Specific requests to FDA

1. Accept this nomination as a high-priority Scenario 1 candidate under the Drug Repurposing RFI.
2. Publish a transparent evidence-review protocol and evidence table addressing the exact regimen, patient population, endpoints, counseling cointerventions, and safety safeguards.
3. Evaluate whether the direct targeted trials plus confirmatory daily-dose, mechanistic, and external-regulatory evidence satisfy §505(d), including the one-AWC-investigation-plus-confirmatory-evidence route where legally appropriate.
4. Confirm ReVia's eligibility under each §503D covered-drug predicate and, if eligible and evidence supports a change, initiate MODERN notice and labeling procedures.
5. Use the anthrax and decorporation precedents as process models for agency-led literature curation, while applying current statutory requirements and indication-specific standards.
6. If a residual gap remains, articulate it precisely and facilitate an efficient pragmatic randomized/RWE package rather than defaulting to a conventional sponsor model that generic economics cannot support.
7. Engage generic manufacturers, investigators, health systems, telehealth programs, payers, EHR/clinical-decision-support vendors, nonprofits, patient communities, and Curada in a neutral evidence-development consortium.
8. Ensure any future communication clearly distinguishes approved daily naltrexone, proposed targeted naltrexone, and EU-authorized as-needed nalmeferene.

Curada's commitment

Curada stands ready to assist FDA with literature curation, protocol development, data-standard alignment, stakeholder coordination, and prospective real-world evidence generation. Curada supports a scientifically conservative process: transparent methods, corrected citations, prespecified analyses, full reporting of unfavorable findings, and no substitution of anecdotes or utilization figures for the statutory evidence standard.

Conclusion

Targeted oral naltrexone is exactly the kind of repurposing opportunity the RFI is designed to surface: a familiar, inexpensive, orally administered medicine; a major undertreated condition; a plausible and clinically tested alternative regimen; an unusually long public research history; a withdrawn RLD not withdrawn for safety or effectiveness; and weak commercial incentives for label modernization. The evidence is substantial enough to warrant immediate FDA review and focused enough to make that review tractable. The appropriate public-health response is neither uncritical endorsement nor continued neglect—it is a rigorous, transparent determination followed by labeling action or a narrowly tailored confirmatory pathway.

References and source links

All web links accessed July 13, 2026. PubMed identifiers and trial descriptions were checked against primary records. Official FDA, Federal Register, statute, EMA, and NICE sources are prioritized.

1. FDA. Drug Repurposing for Unmet Medical Needs; Request for Information. 91 Fed. Reg. 25897 (May 12, 2026). [Source](#)
2. FDA. Drug Repurposing for Unmet Medical Needs; Extension of Comment Period. 91 Fed. Reg. 35488 (June 11, 2026). [Source](#)
3. FDA. Determination That ReVia (Naltrexone Hydrochloride) Tablets, 50 Milligrams, Was Not Withdrawn From Sale for Reasons of Safety or Effectiveness. 91 Fed. Reg. 20471 (Apr. 16, 2026). [Source](#)
4. 21 U.S.C. §353d (FD&C Act §503D), Modernizing Outdated Labeling for Generic Drugs Act (MODERN). [Source](#)
5. FDA. MODERN statutory text / Consolidated Appropriations Act, 2021, §324 (FDA legislative compilation). [Source](#)
6. DailyMed. Naltrexone hydrochloride tablets, prescribing information (mechanism, pharmacokinetics, indications, contraindications, warnings). [Source](#)
7. Spanagel R, et al. The neuronal circuits of addiction: alcohol and opioid pathways / mechanistic review. [Source](#)
8. Ray LA, Hutchison KE. Effects of naltrexone on alcohol sensitivity and genetic moderation (mechanistic context). [Source](#)
9. Weerts EM, et al. PET evidence concerning naltrexone opioid-receptor blockade. PMID 17487229. [Source](#)
10. Sinclair JD. Evidence about the use of naltrexone and for different ways of using it in the treatment of alcoholism. Alcohol Alcohol. 2001;36:2–10. PMID 11139409. [Source](#)
11. Kranzler HR, et al. Targeted naltrexone treatment of early problem drinkers. Addict Behav. 1997;22:431–436. PMID 9183513; doi:10.1016/S0306-4603(96)00064-0. [Source](#)
12. Heinälä P, et al. Targeted use of naltrexone without prior detoxification in the treatment of alcohol dependence. J Clin Psychopharmacol. 2001;21:287–292. PMID 11386491; doi:10.1097/00004714-200106000-00006. [Source](#)
13. Kranzler HR, et al. Targeted naltrexone for early problem drinkers. J Clin Psychopharmacol. 2003;23:294–304. PMID 12826991; doi:10.1097/01.jcp.0000084030.22282.6d. [Source](#)
14. Hernandez-Avila CA, et al. Targeted versus daily naltrexone: secondary analysis. Alcohol Clin Exp Res. 2006;30:860–865. PMID 16634855; doi:10.1111/j.1530-0277.2006.00101.x. [Source](#)
15. Kranzler HR, et al. Targeted naltrexone for problem drinkers. J Clin Psychopharmacol. 2009;29:350–357. PMID 19593174; PMCID PMC3184851. [Source](#)
16. O'Malley SS, et al. Reduction of alcohol drinking in young adults by naltrexone. J Clin Psychiatry. 2015;76:e207–e213. PMID 25742208. [Source](#)
17. Santos GM, et al. Targeted oral naltrexone for mild to moderate AUD among sexual and gender minority men. Am J Psychiatry. 2022;179:915–926. PMID 36285404; doi:10.1176/appi.ajp.20220335. [Source](#)
18. Anton RF, et al. Combined pharmacotherapies and behavioral interventions for alcohol dependence: the COMBINE study. JAMA. 2006;295:2003–2017. PMID 16670409; doi:10.1001/jama.295.17.2003. [Source](#)
19. Sinclair JD. U.S. Patent 4,882,335, Method for treating alcohol-drinking response (filed 1988; issued 1989; expired). [Source](#)
20. European Medicines Agency. Selincro (nalmefene), European Public Assessment Report. [Source](#)
21. European Medicines Agency. Selincro product information (as-needed dosing and indication). [Source](#)
22. NICE. Technology appraisal guidance TA325: Nalmefene for reducing alcohol consumption in people with alcohol dependence. [Source](#)
23. NICE. Clinical guideline CG115: Alcohol-use disorders—diagnosis, assessment and management; oral naltrexone recommendations. [Source](#)
24. NIAAA. Alcohol Treatment in the United States (2024 NSDUH estimates; page updated March 2026). [Source](#)
25. CDC. Excessive Alcohol Use–Related Deaths and Years of Potential Life Lost—United States, 2020–2021. MMWR. 2024;73. [Source](#)
26. Williams C, Boyer D, Carino T. Regulatory Opportunities to Advance Generic Drug Repurposing. Duke-Margolis Institute for Health Policy. Sept. 5, 2025. [Source](#)
27. Options Save Lives. About the nonprofit and Sinclair Method mission. [Source](#)
28. Thrive Alcohol Recovery. Sinclair Method coaching and community. [Source](#)
29. Oar Health. Sinclair Method / event-driven naltrexone information; provider-reported membership. [Source](#)
30. Ria Health. Patient education describing daily or pre-drinking naltrexone dosing. [Source](#)
31. Ria Health. Research summary and cohort description. [Source](#)
32. Choose Your Horizon. Telehealth AUD treatment and coaching. [Source](#)
33. Sunnyside. Educational discussion of the Sinclair Method and its separate daily-adherence protocol. [Source](#)
34. Glaser G. The Irrationality of Alcoholics Anonymous. The Atlantic. April 2015. [Source](#)
35. Harvard Health Publishing. Can you retrain your brain to stop excessive drinking? 2026. [Source](#)
36. WNYC/NPR-WBUR. Public-radio discussion of naltrexone and drinking reduction. [Source](#)
37. Partnership to End Addiction. The Sinclair Method: a medication-assisted treatment for alcohol use disorder. [Source](#)
38. FDA. Orange Book Data Files. [Source](#)
39. FDA. Real-World Evidence Program. [Source](#)
40. FDA. Doxycycline and Penicillin G Procaine for Inhalational Anthrax (Post-Exposure). [Source](#)

41. FDA. Prescription Drug Products; Doxycycline and Penicillin G Procaine Administration for Inhalational Anthrax. 66 Fed. Reg. 55679 (Nov. 2, 2001). [Source](#)
42. FDA. Guidance: Calcium-DTPA and Zinc-DTPA Drug Products—Submitting a New Drug Application. [Source](#)
43. FDA. Approval documents, Pentetate Calcium Trisodium and Pentetate Zinc Trisodium (2004). [Source](#)
44. FDA. FDA Approves Drugs to Treat Internal Contamination From Radioactive Elements. [Source](#)
45. McPheeters M, et al. Pharmacotherapy for Alcohol Use Disorder: a systematic review and meta-analysis. JAMA. 2023;330:1653–1665. PMID PMC10630900. [Source](#)
46. Australian Therapeutic Goods Administration. ARTG entry for naltrexone hydrochloride 50-mg tablets. [Source](#)
47. Australian Guidelines for the Treatment of Alcohol Problems. Naltrexone. [Source](#)
48. Grant JE, et al. Multicenter investigation of naltrexone in pathological gambling. J Clin Psychiatry. 2008. PMID 18384246. [Source](#)
49. Savard J, et al. Naltrexone in compulsive sexual behavior: open-label feasibility study. PMID 32532705. [Source](#)
50. Lew-Starowicz M, et al. Tolerability and efficacy of paroxetine and naltrexone for compulsive sexual behavior disorder. World Psychiatry. 2022;21:321–322. [Source](#)
51. Schmitz JM, et al. Naltrexone and relapse prevention treatment for cocaine-dependent patients. PMID 11316375. [Source](#)
52. O'Malley SS, et al. Naltrexone decreases craving and alcohol self-administration in alcohol-dependent subjects and activates the hypothalamo-pituitary-adrenocortical axis. Psychopharmacology. 2002;160:19–29. PMID 11862370; doi:10.1007/s002130100919. [Source](#)
53. Rösner S, et al. Opioid antagonists for alcohol dependence. Cochrane Database Syst Rev. 2010;(12):CD001867. PMID 21154349; doi:10.1002/14651858.CD001867.pub3. [Source](#)
54. Palpacuer C, et al. Pharmacologically controlled drinking in the treatment of alcohol dependence or alcohol use disorders: a systematic review with direct and network meta-analyses. Addiction. 2018;113:220–237. PMID 28940866; doi:10.1111/add.13974. [Source](#)
55. FDA. Qualification of a two-level reduction in alcohol risk-drinking level as an acceptable primary endpoint for trials in adults with moderate-to-severe AUD. Feb. 14, 2025. [Source](#)
56. FDA. FDA Approves First Treatment for Patients with Cerebral Folate Transport Deficiency (Wellcovorin label expansion based on systematic literature review). Mar. 10, 2026. [Source](#)

Appendix A. Web comment (under 500 words)

Curada Public Benefit Corporation respectfully submits this comment and the attached evidence dossier in response to FDA's Drug Repurposing for Unmet Medical Needs RFI. We nominate oral naltrexone hydrochloride 50-mg tablets, used on an event-driven schedule approximately 60 minutes before anticipated drinking (with FDA to determine the supported timing window), as a high-priority Scenario 1 candidate for FDA evaluation in alcohol use disorder (AUD). We ask FDA to determine whether the public record satisfies the substantial-evidence standard and, if a narrow gap remains, define an efficient Scenario 2 confirmatory pathway.

The nomination is unusually tractable. Naltrexone is already approved for alcohol dependence. The proposed condition of use retains the same active ingredient, strength, immediate-release tablet, and oral route; the principal questions concern regimen, population, timing, and a reduction-of-heavy-drinking claim. FDA determined in April 2026 that ReVia (NDA 018932) was not withdrawn for reasons of safety or effectiveness. That finding does not validate targeted dosing, but it is relevant to generic reliance and potential eligibility under FD&C Act §503D (MODERN). Prong I—new scientific evidence not reflected in labeling—is the clearest basis; FDA should also assess all covered-drug predicates and whether prong III applies.

The mechanism is plausible but should not be overstated. Naltrexone antagonizes opioid receptors. Administration before alcohol exposure can attenuate endogenous-opioid reinforcement and downstream reward signaling; repeated pairing of alcohol-related cues and consumption with reduced reinforcement may weaken learned reward value, a model often called “pharmacological extinction.” FDA labeling correctly states that naltrexone's mechanism in alcoholism is not fully understood, and plasma Tmax does not establish an exact optimal dosing interval.

Multiple randomized studies directly evaluated targeted or event-driven naltrexone, including trials by Heinälä, Kranzler and colleagues, and Santos and colleagues. Findings are promising but heterogeneous across populations, instructions, counseling, and endpoints. Broader evidence establishes the molecule's AUD activity: a Cochrane review of 50 randomized trials (7,793 participants) found reduced risk of heavy drinking, while later controlled-drinking reviews identify uncertainty. Human laboratory evidence also supports attenuation of alcohol reinforcement, but not the exact event-driven schedule. Europe supplies a class-and-regimen precedent: EMA authorized as-needed nalmefene, and NICE TA325 recommends it for selected patients. NICE has not endorsed targeted naltrexone.

The labeling gap reflects structural market failure. The relevant method patent expired; any entity funding label development would create a benefit available to generic competitors. Duke-Margolis has described resulting reimbursement, awareness, uptake, and liability barriers. Provider and community reach supports feasibility, not effectiveness.

Curada offers to support a protocol-driven literature synthesis and a prospective, auditable evidence consortium. A fit-for-purpose program could capture dose and first-drink timestamps, the FDA-qualified two-level WHO risk-drinking reduction endpoint where applicable, drinking intensity, biomarkers, safety, and prespecified comparator analyses. FDA should prioritize this nomination, initiate §503D or a literature-led labeling process if the evidence supports it, and specify the smallest remaining evidentiary step if it does not.

Submitted by Mariia Markitanova, Founder & Director, Curada Public Benefit Corporation.